

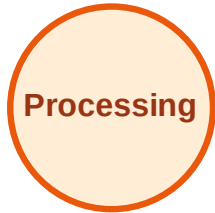
BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Complete



Processing



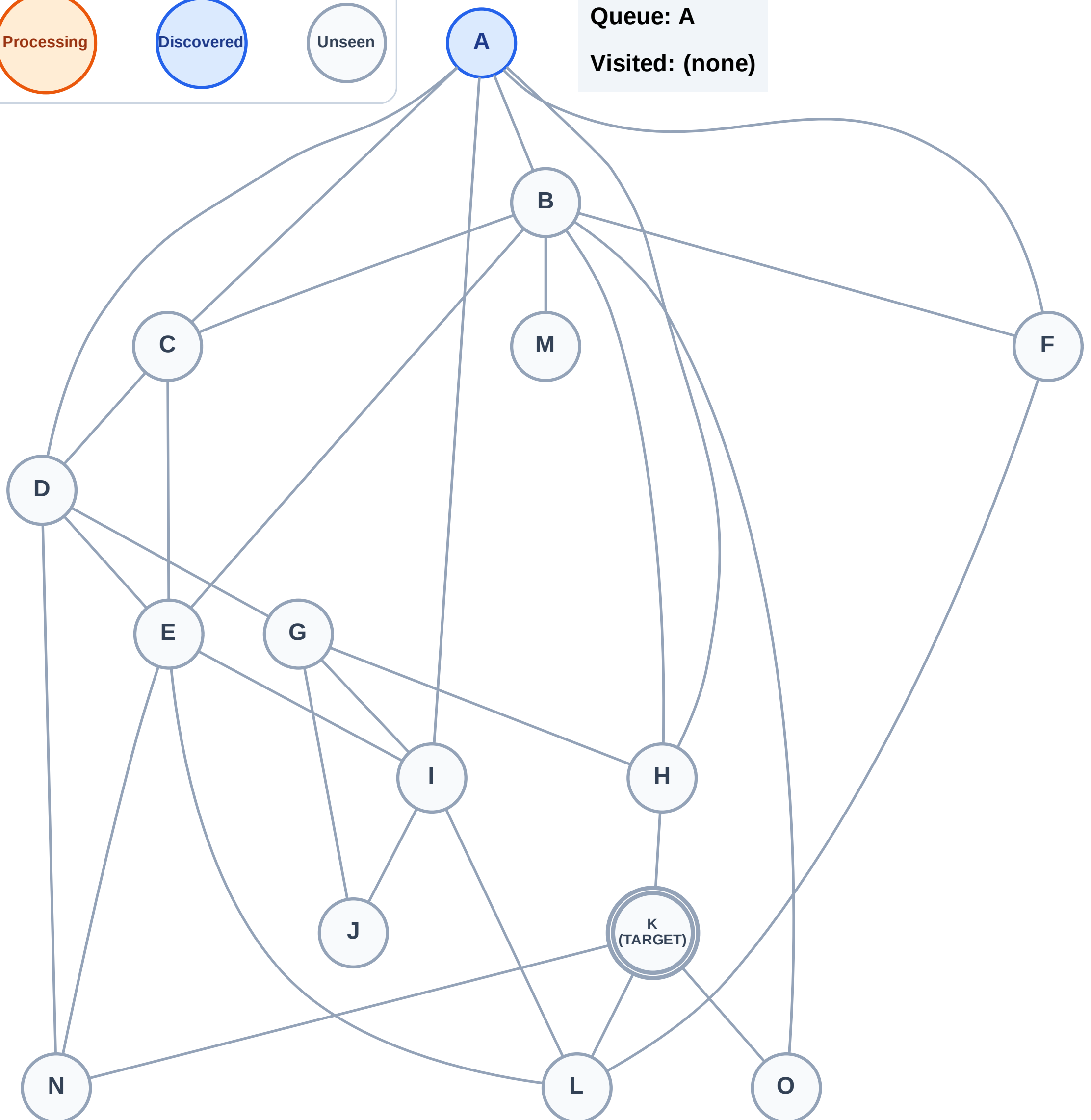
Discovered



Unseen

Queue: A

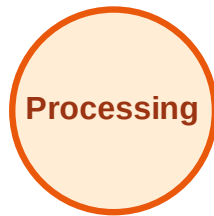
Visited: (none)



BFS Traversal

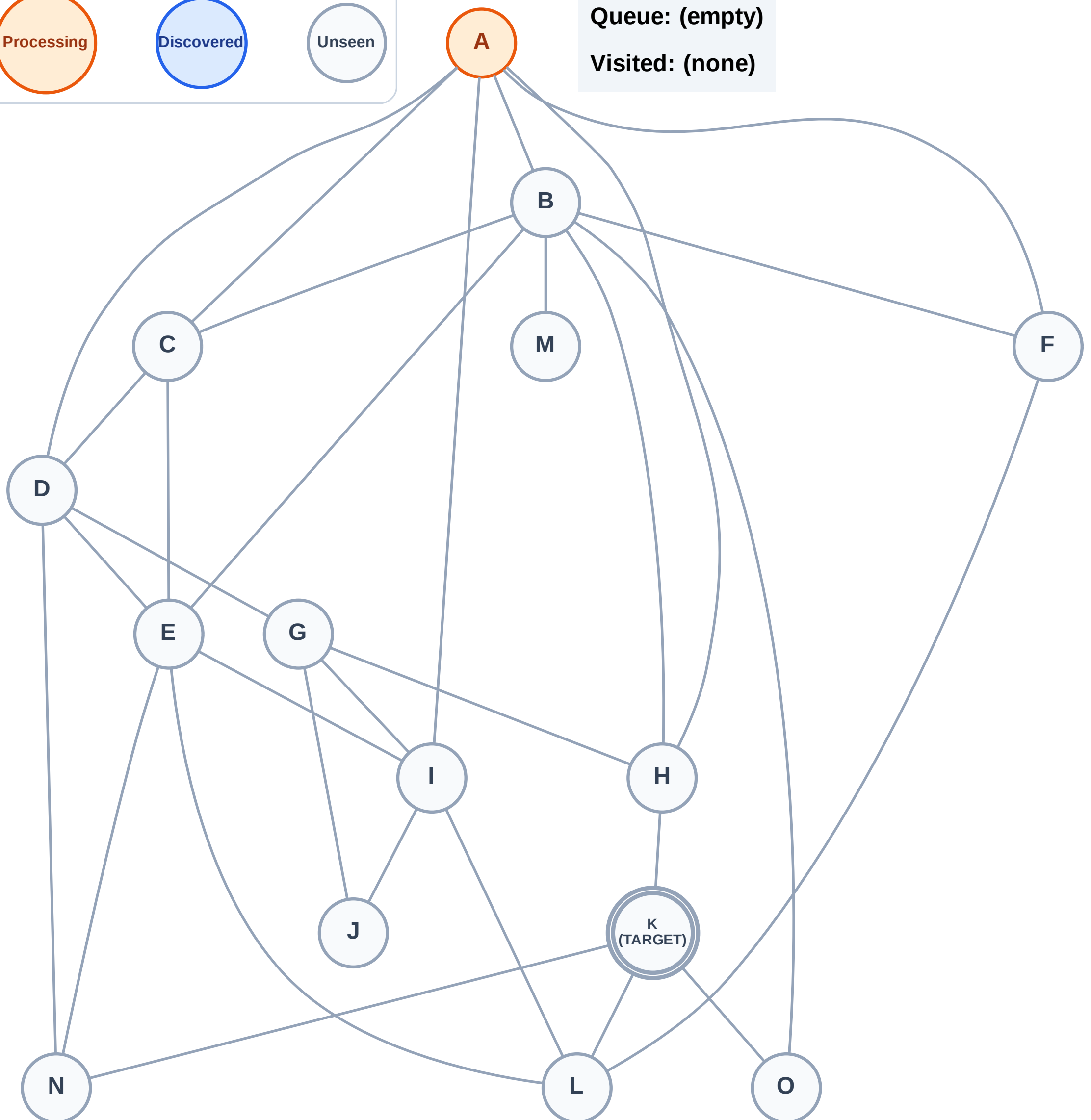
Step 2: Process node A

Legend



Queue: (empty)

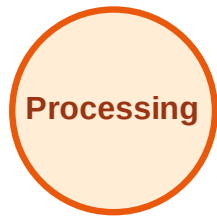
Visited: (none)



BFS Traversal

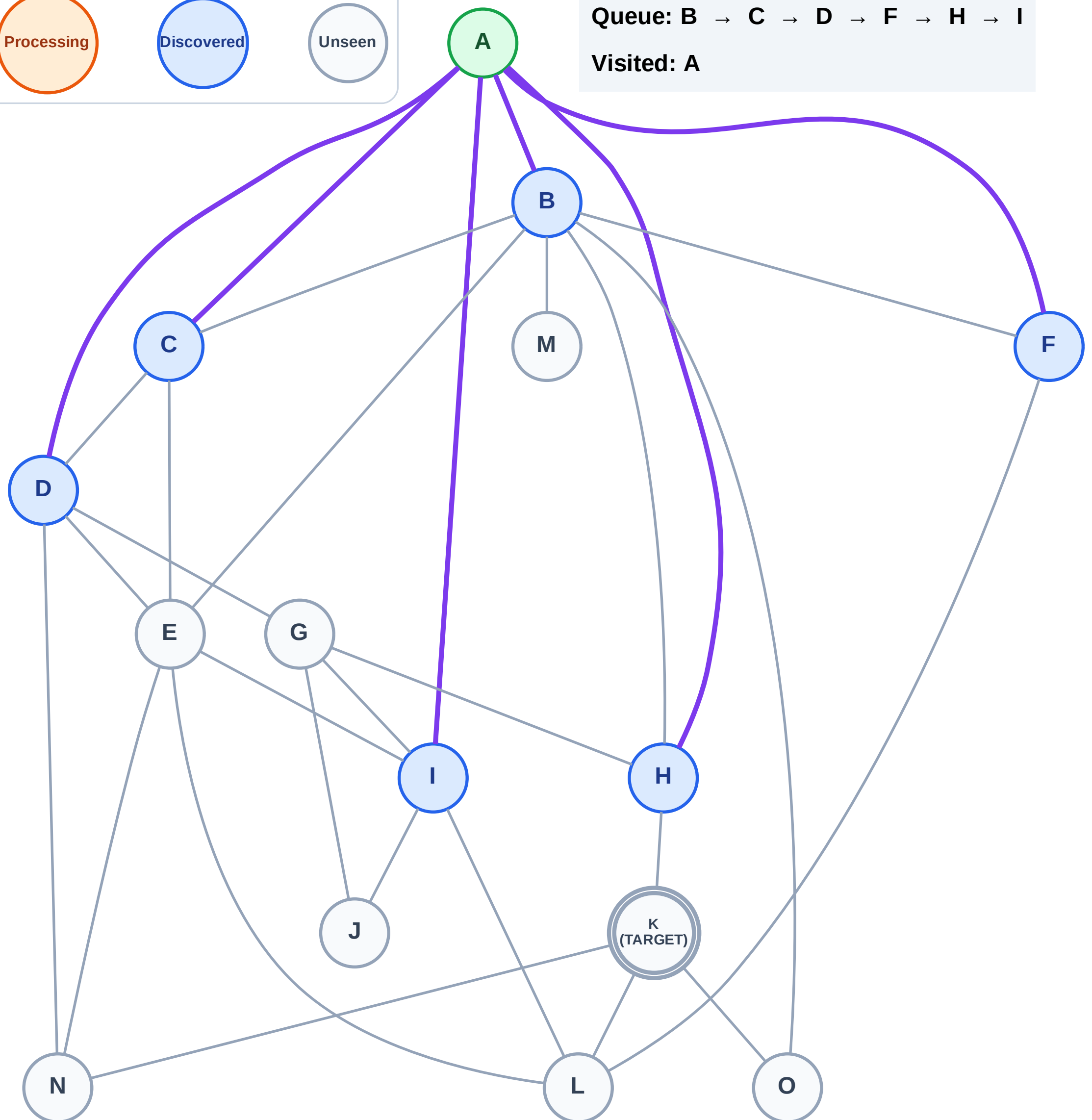
Step 3: Discover B, C, D, F, H, I from A

Legend



Queue: B → C → D → F → H → I

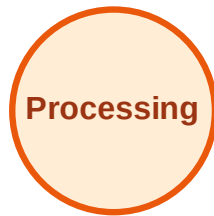
Visited: A



BFS Traversal

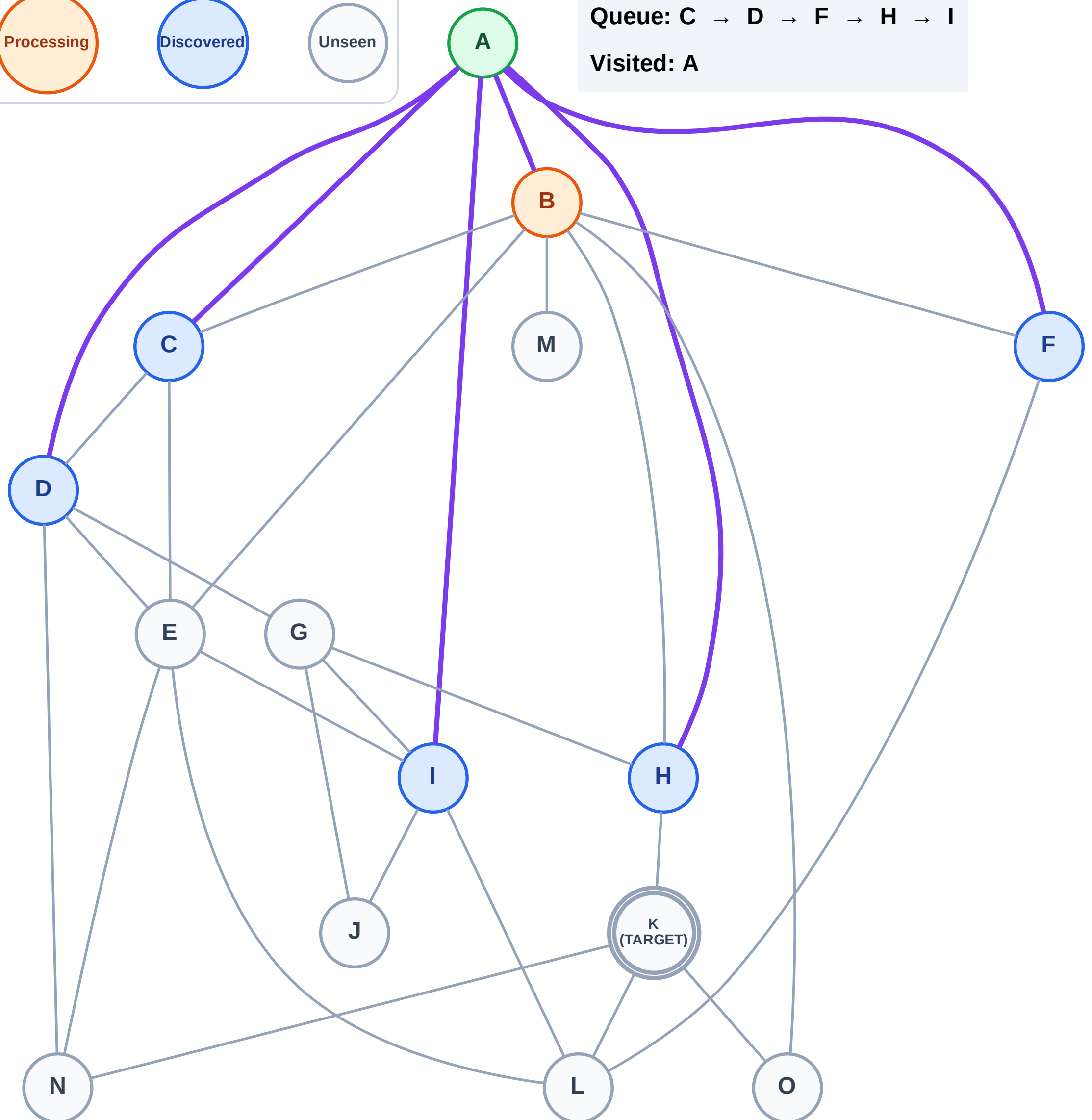
Step 4: Process node B

Legend



Queue: C → D → F → H → I

Visited: A



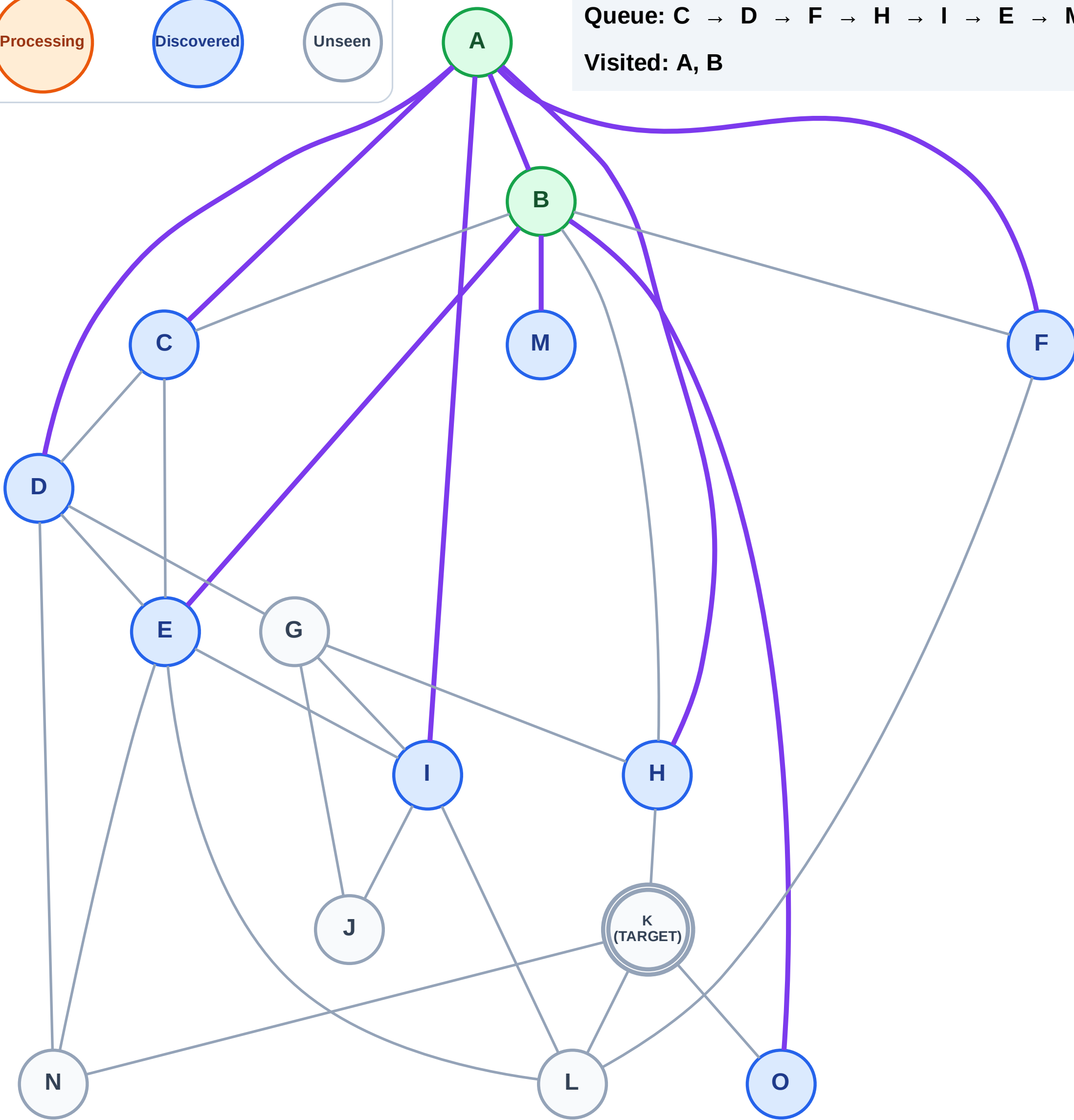
BFS Traversal

Step 5: Discover E, M, O from B

Legend



Queue: C → D → F → H → I → E → M → O
Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

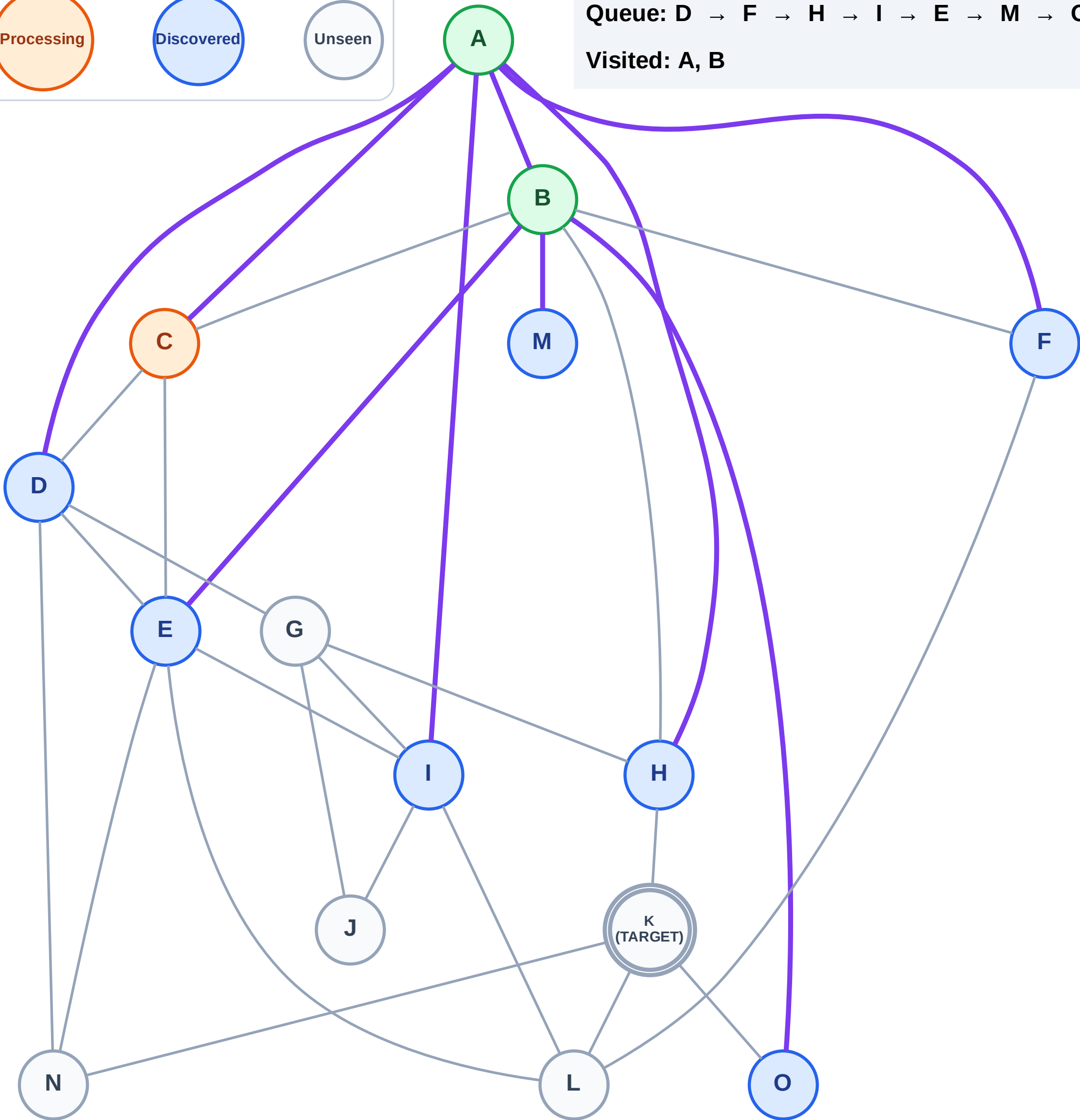
Processing

Discovered

Unseen

Queue: D → F → H → I → E → M → O

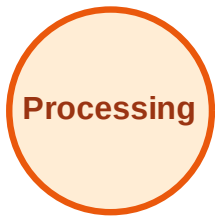
Visited: A, B



BFS Traversal

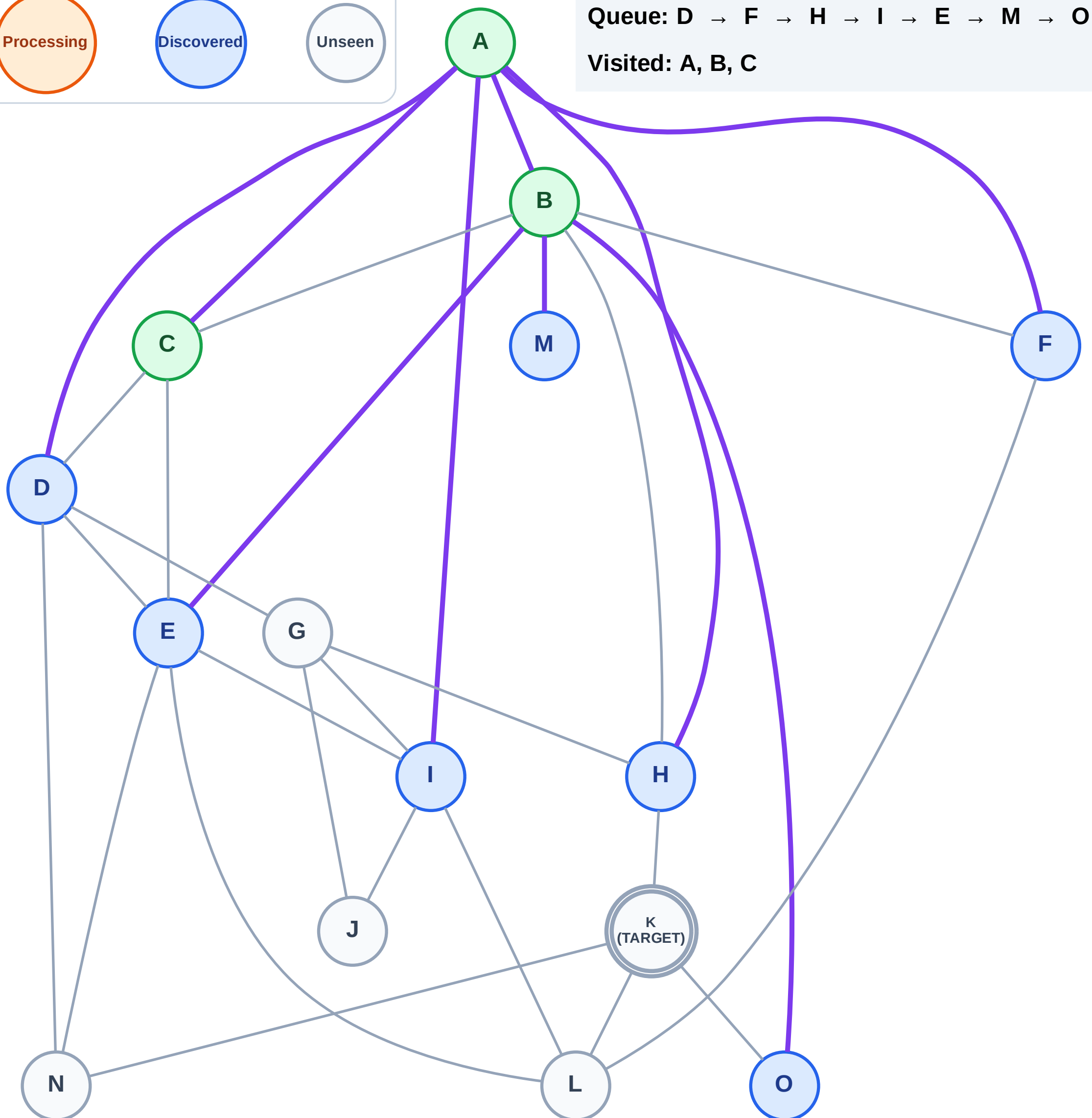
Step 7: No new neighbors from C

Legend



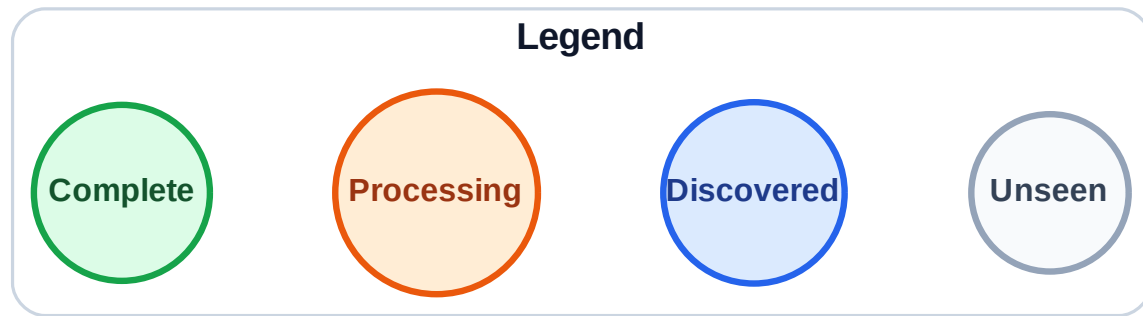
Queue: D → F → H → I → E → M → O

Visited: A, B, C



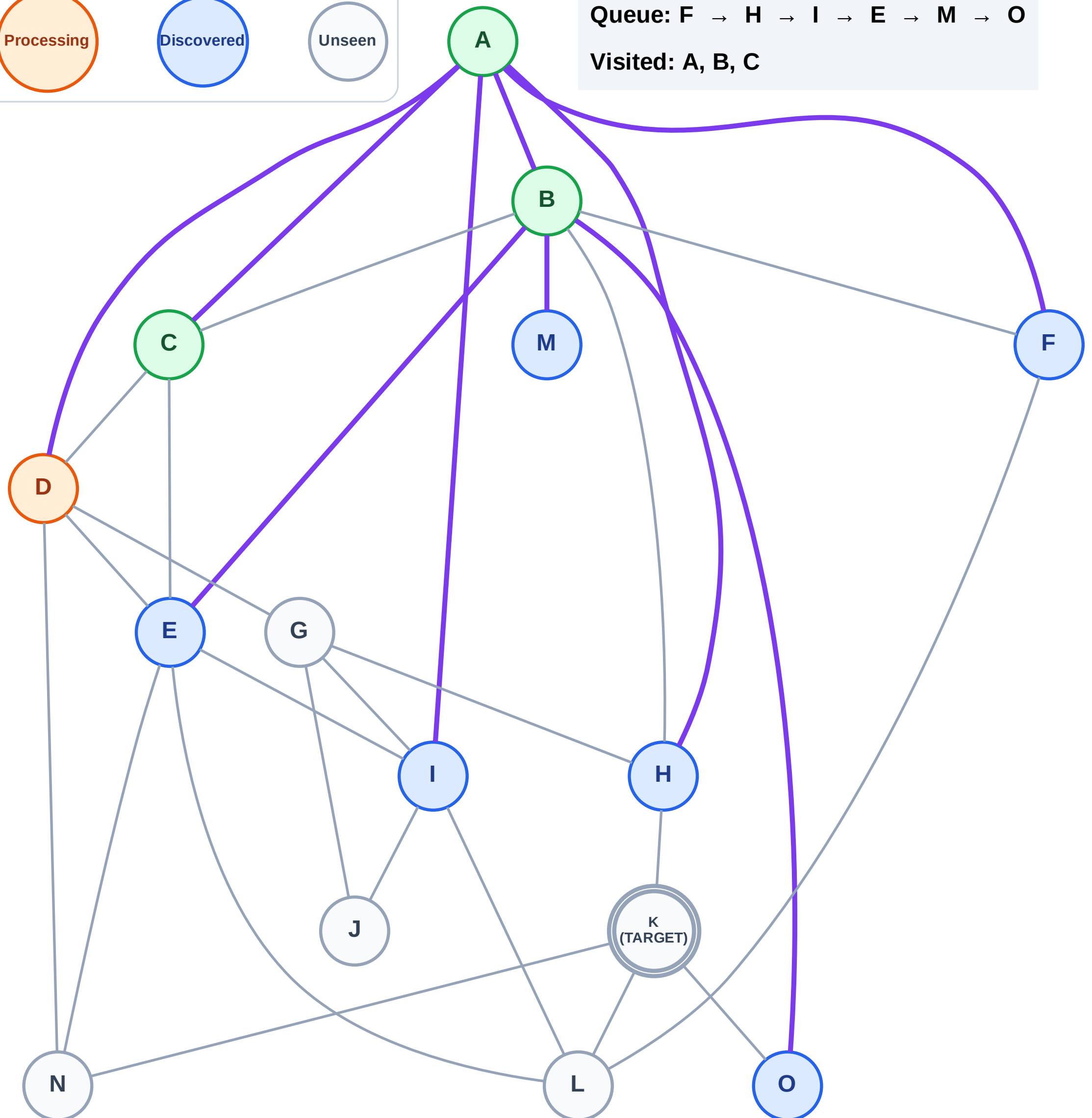
BFS Traversal

Step 8: Process node D



Queue: F → H → I → E → M → O

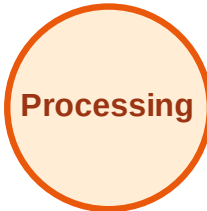
Visited: A, B, C



BFS Traversal

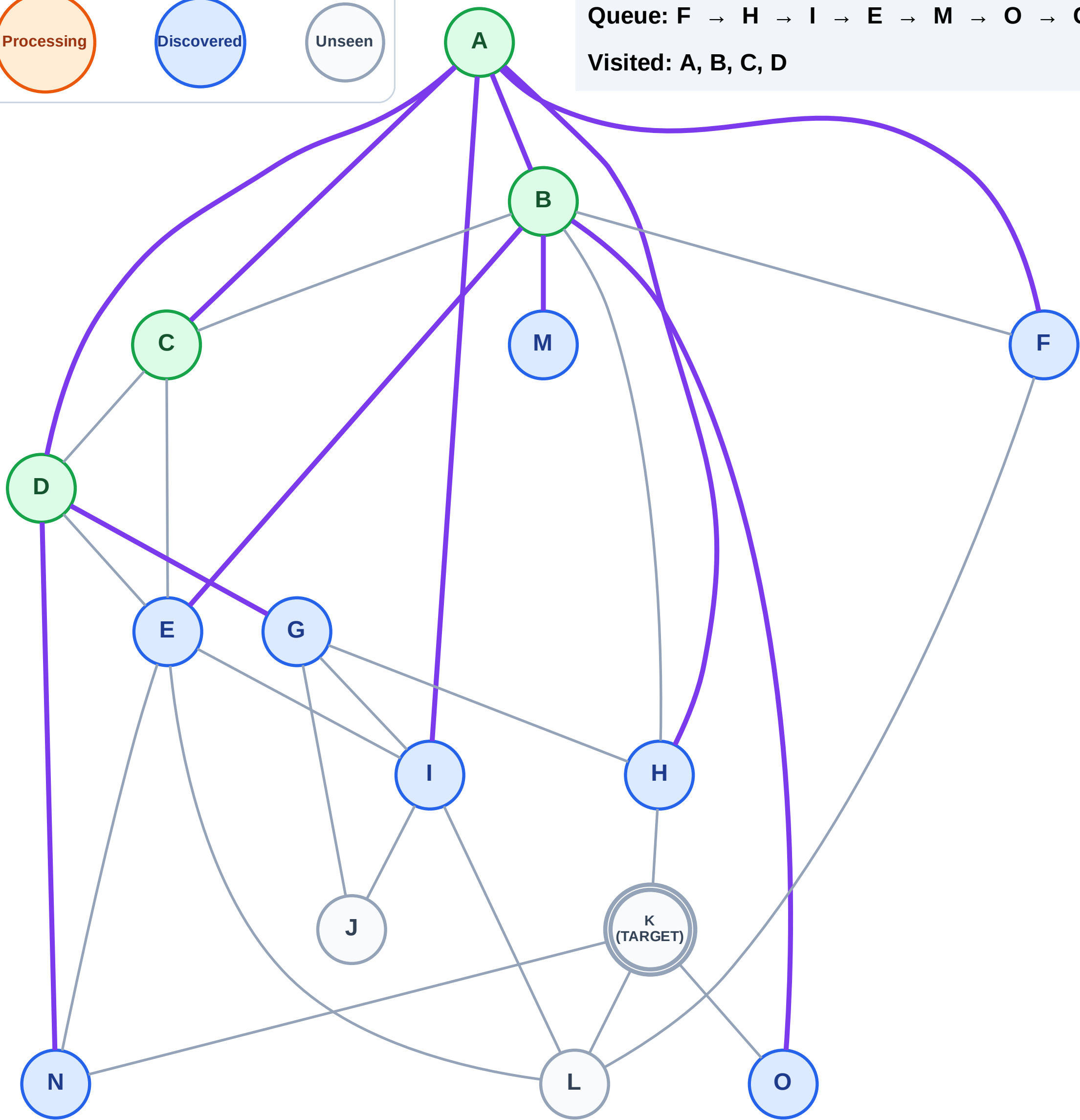
Step 9: Discover G, N from D

Legend



Queue: F → H → I → E → M → O → G → N

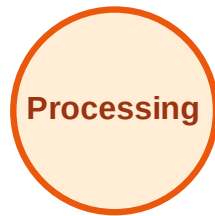
Visited: A, B, C, D



BFS Traversal

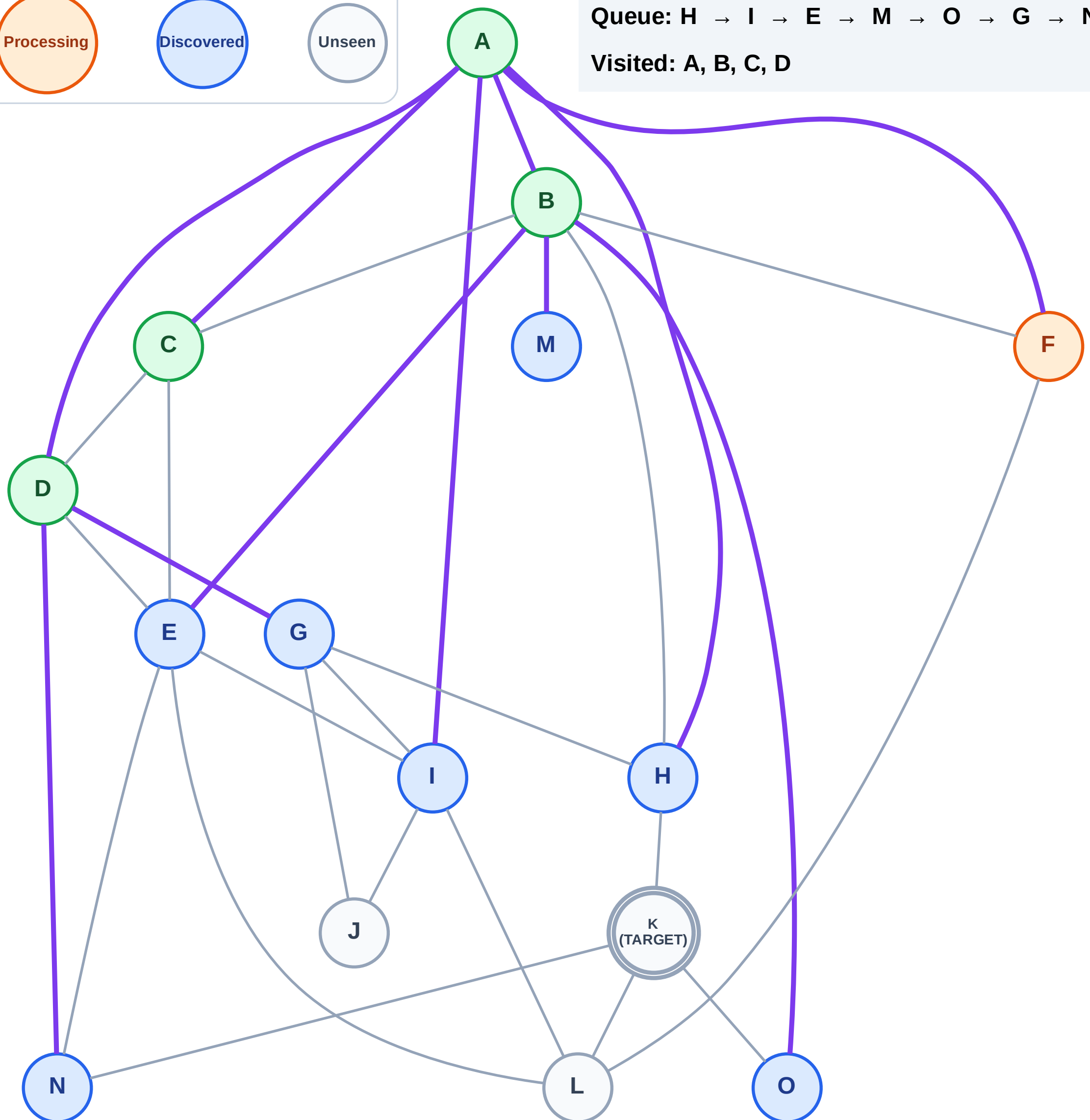
Step 10: Process node F

Legend



Queue: H → I → E → M → O → G → N

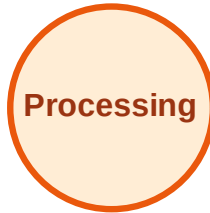
Visited: A, B, C, D



BFS Traversal

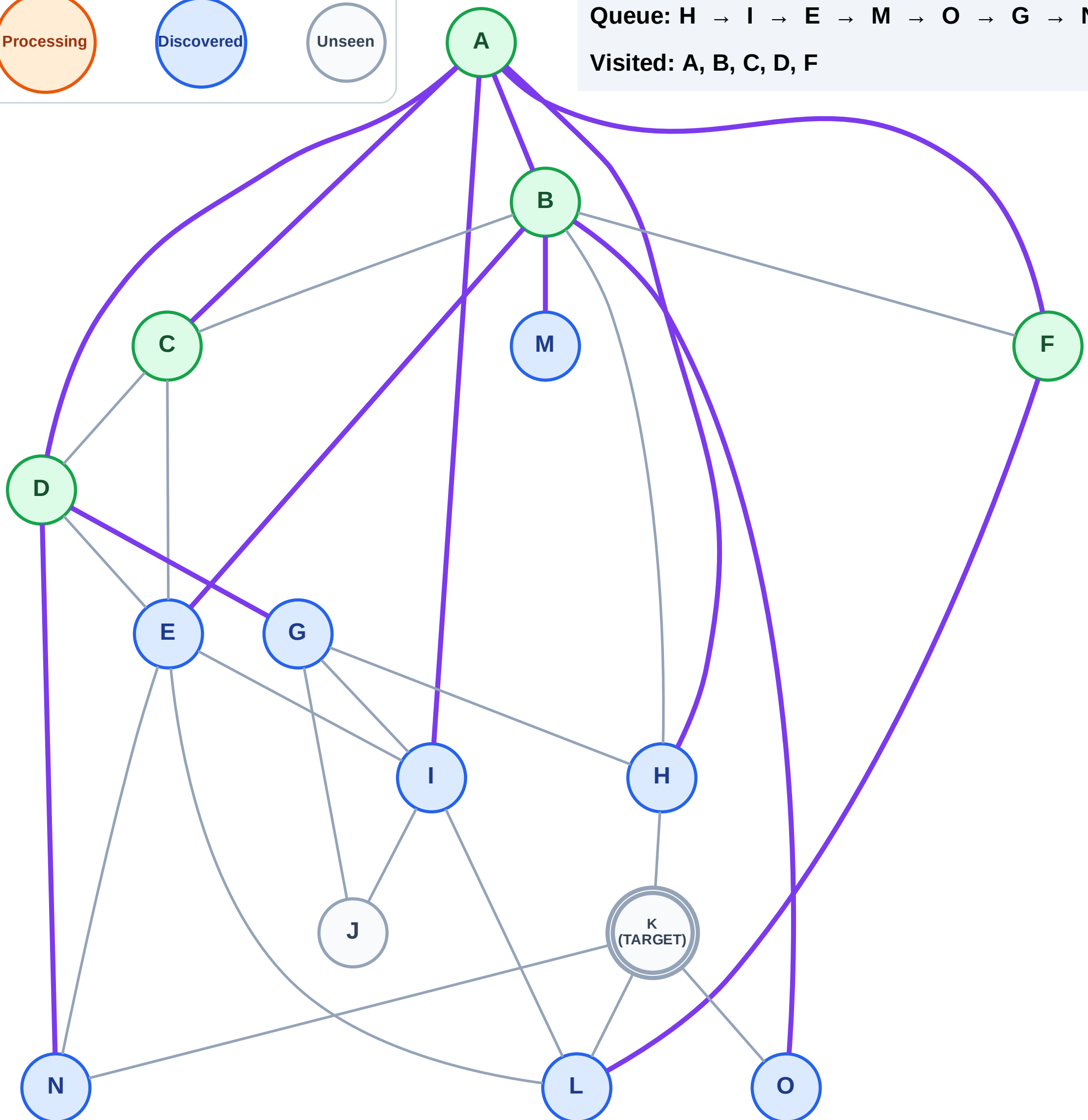
Step 11: Discover L from F

Legend



Queue: H → I → E → M → O → G → N → L

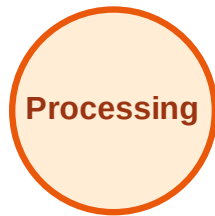
Visited: A, B, C, D, F



BFS Traversal

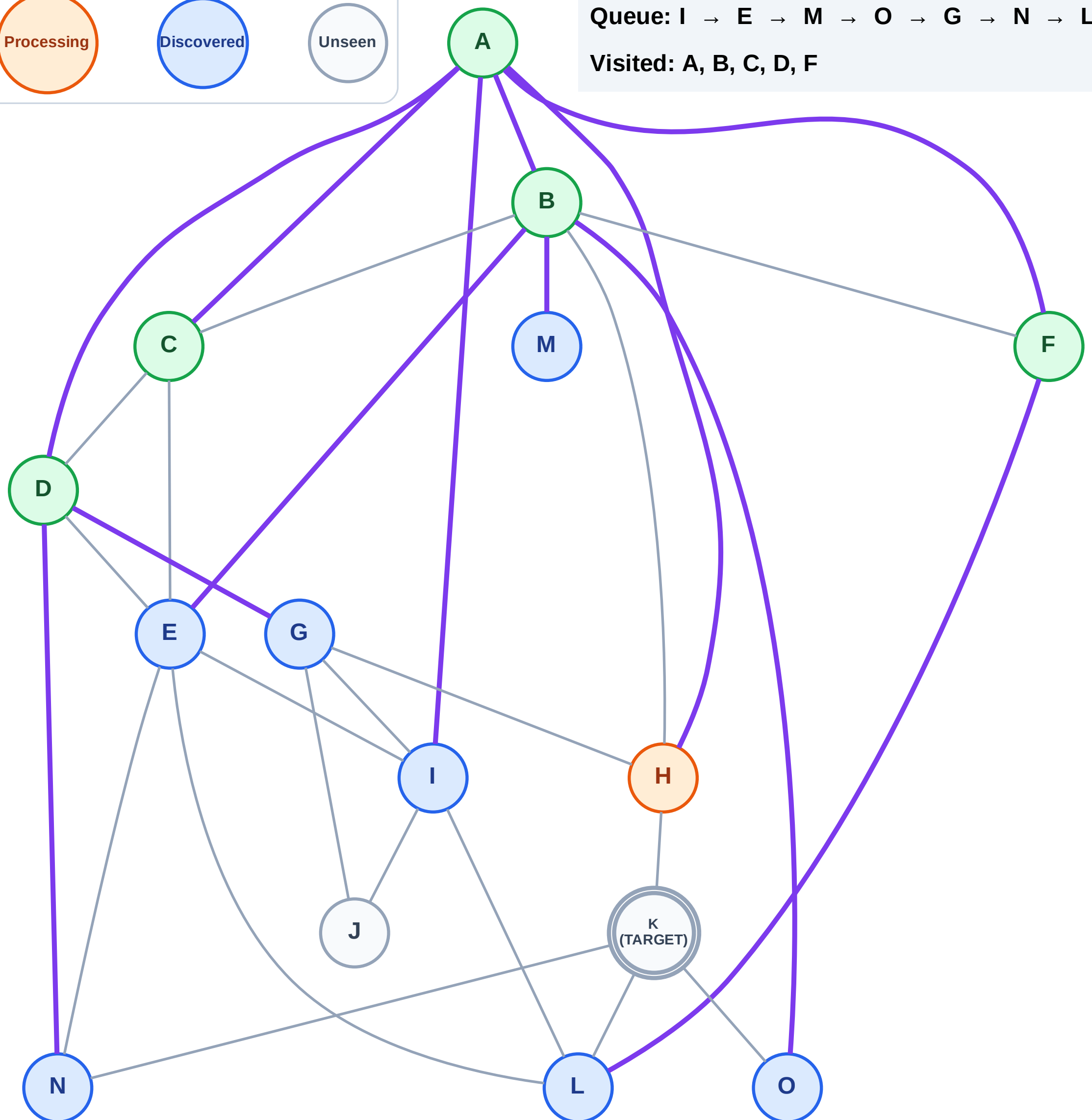
Step 12: Process node H

Legend



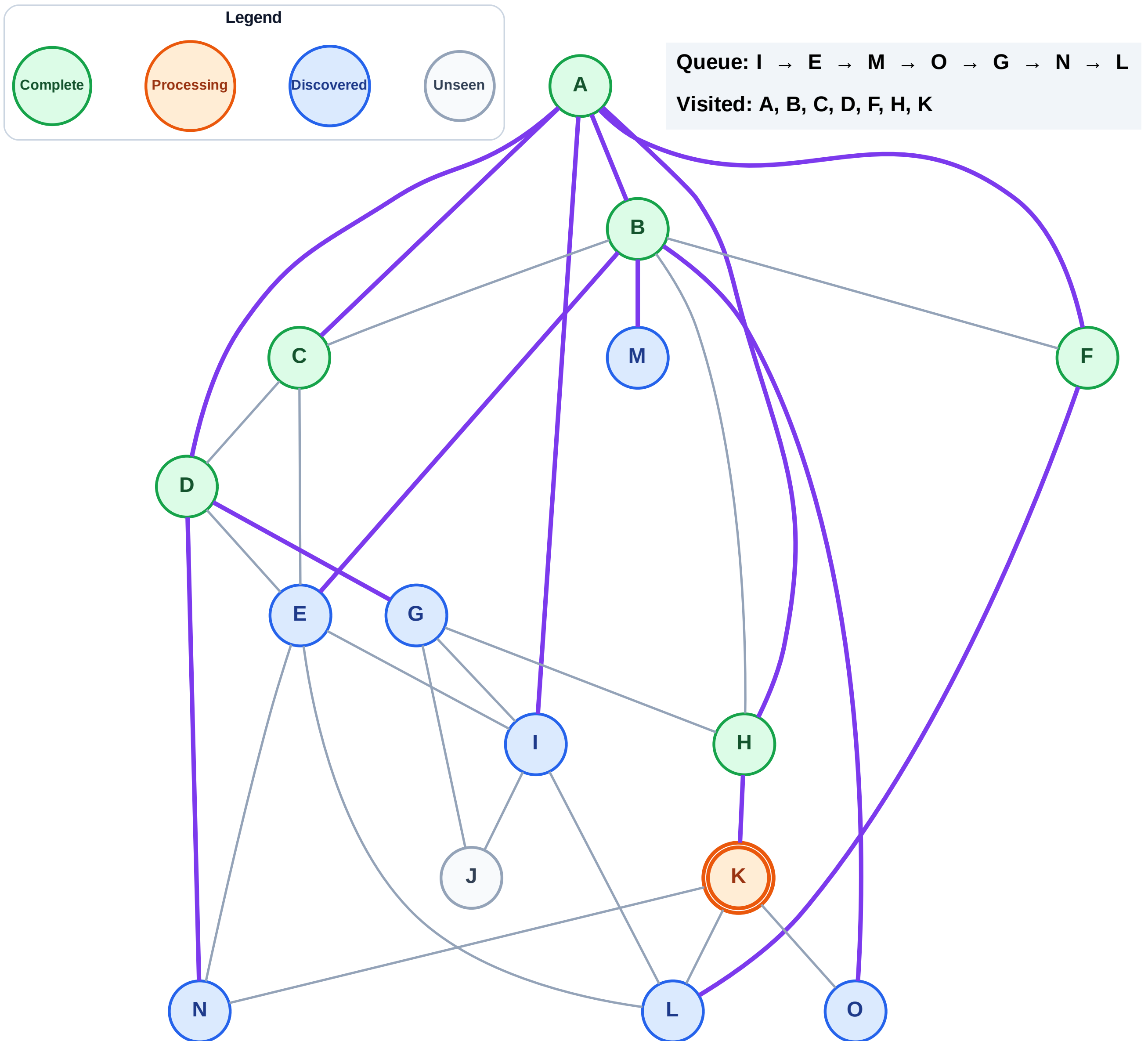
Queue: I → E → M → O → G → N → L

Visited: A, B, C, D, F



BFS Traversal

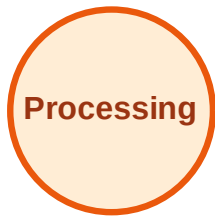
Step 13: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → H → K



DFS Traversal

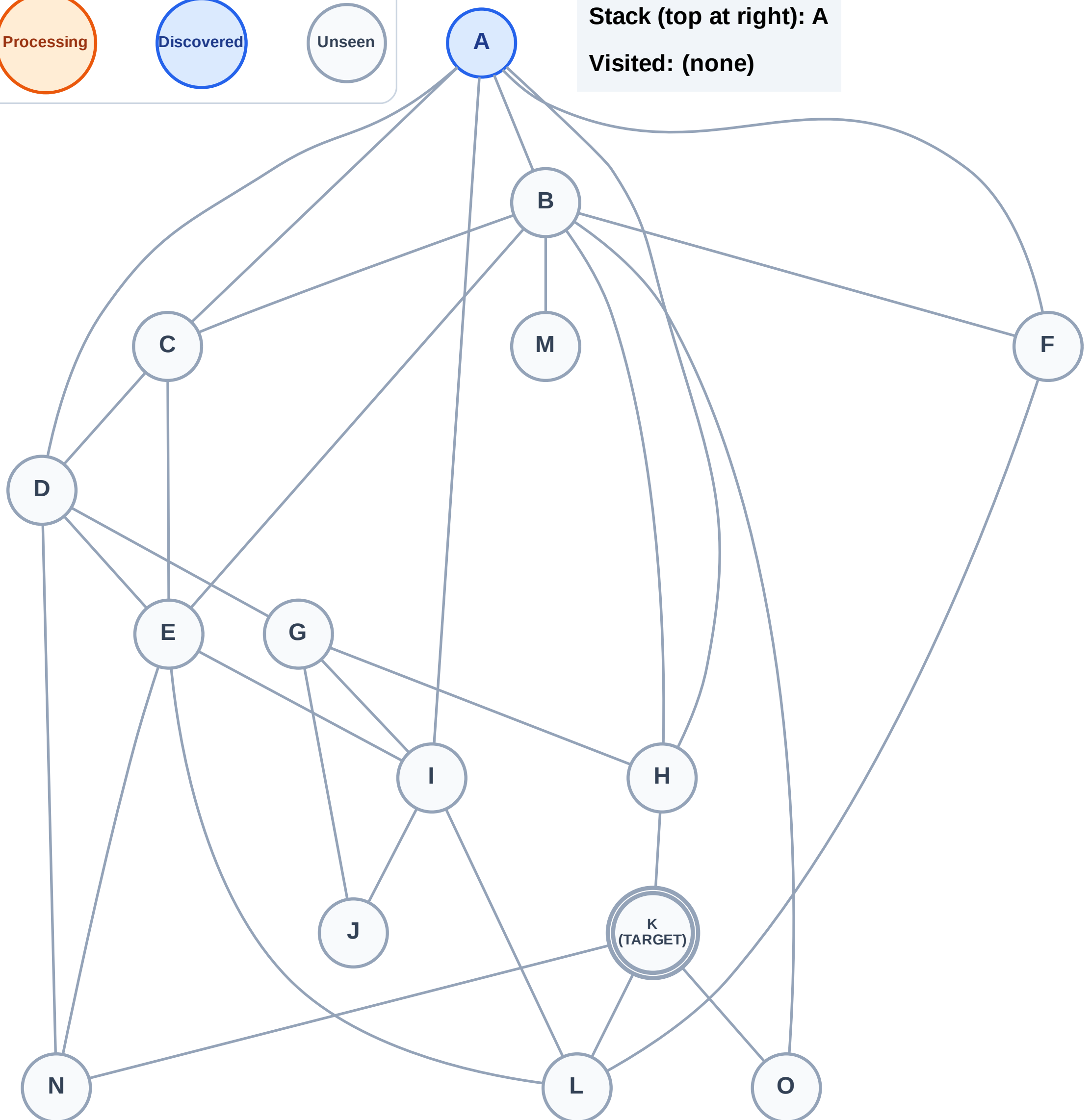
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A

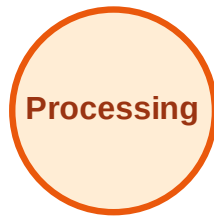
Visited: (none)



DFS Traversal

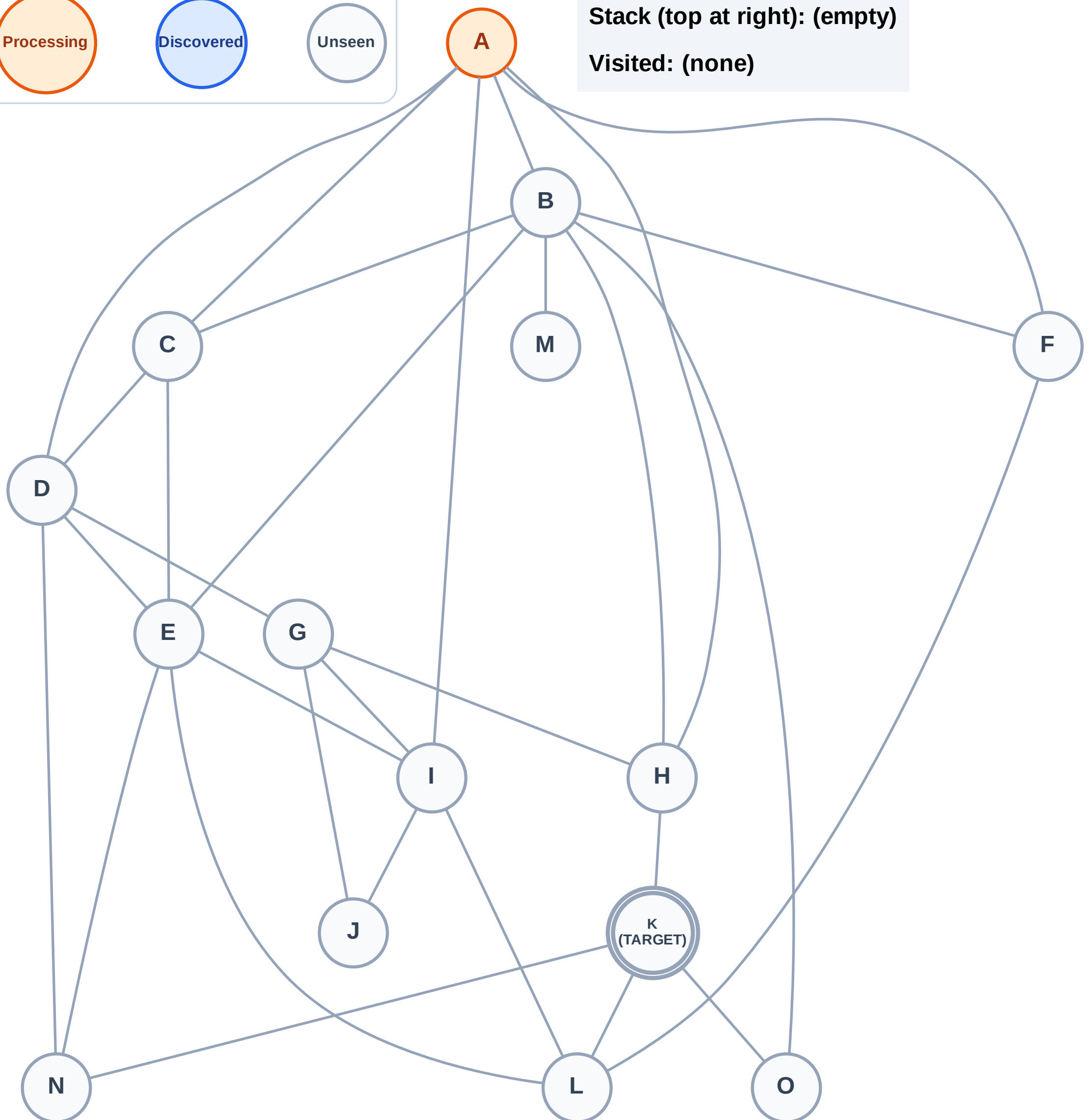
Step 2: Process node A

Legend



Stack (top at right): (empty)

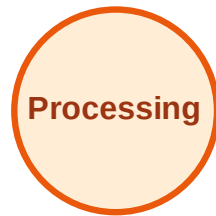
Visited: (none)



DFS Traversal

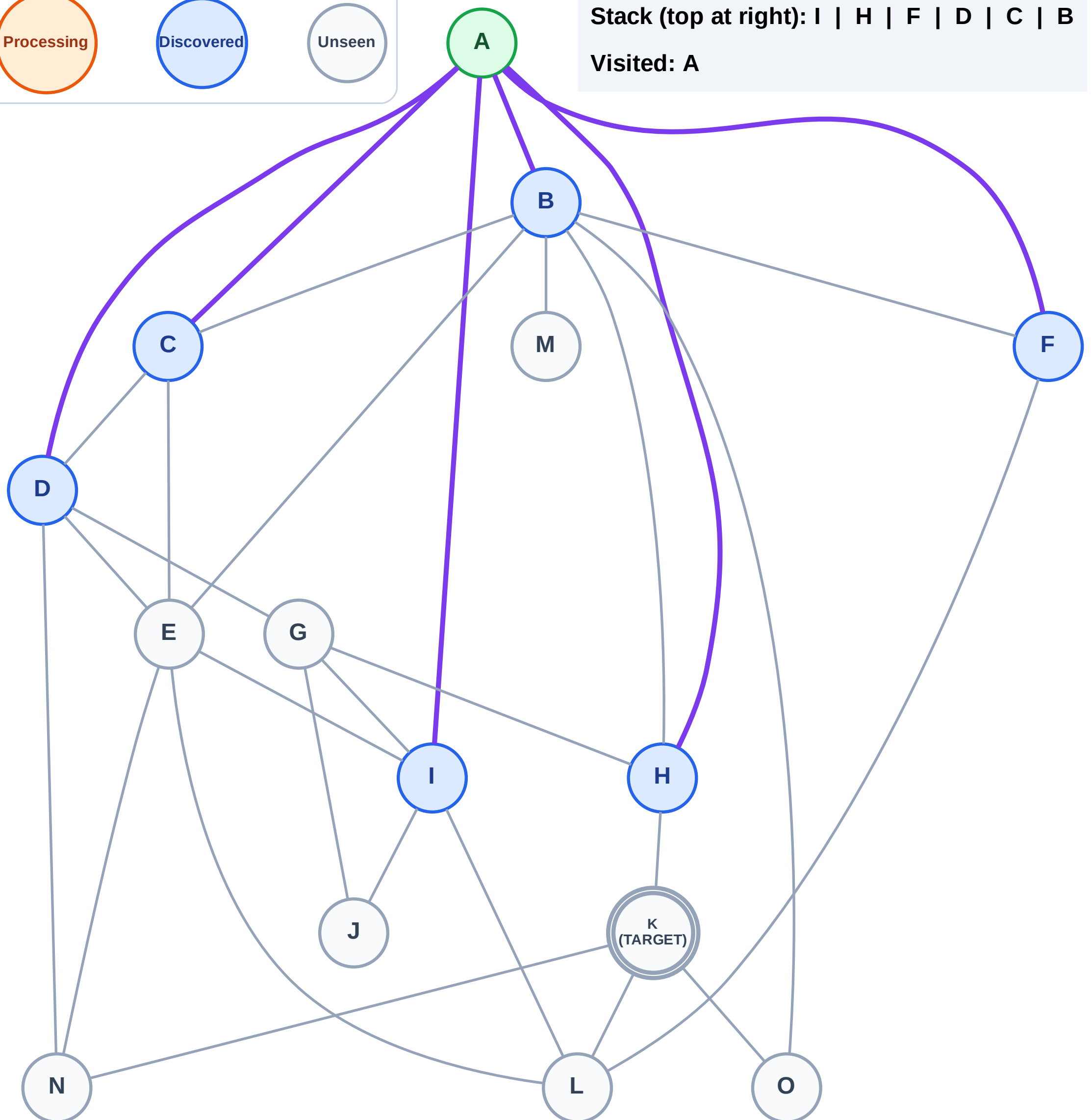
Step 3: Discover I, H, F, D, C, B from A

Legend



Stack (top at right): I | H | F | D | C | B

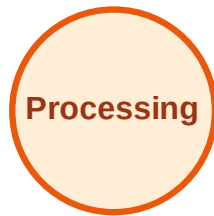
Visited: A



DFS Traversal

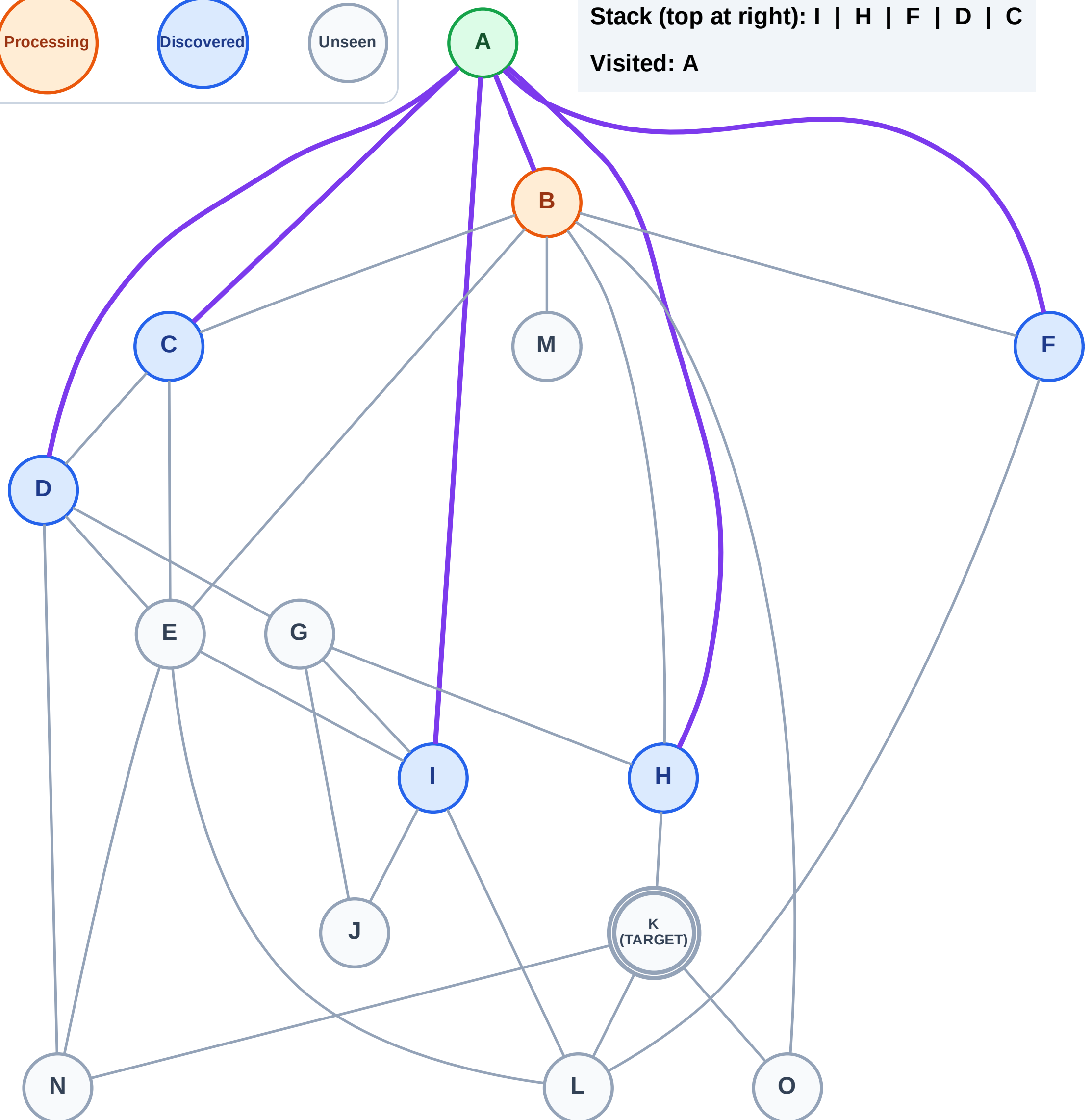
Step 4: Process node B

Legend



Stack (top at right): I | H | F | D | C

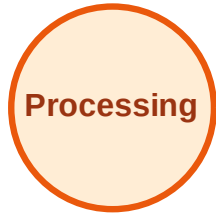
Visited: A



DFS Traversal

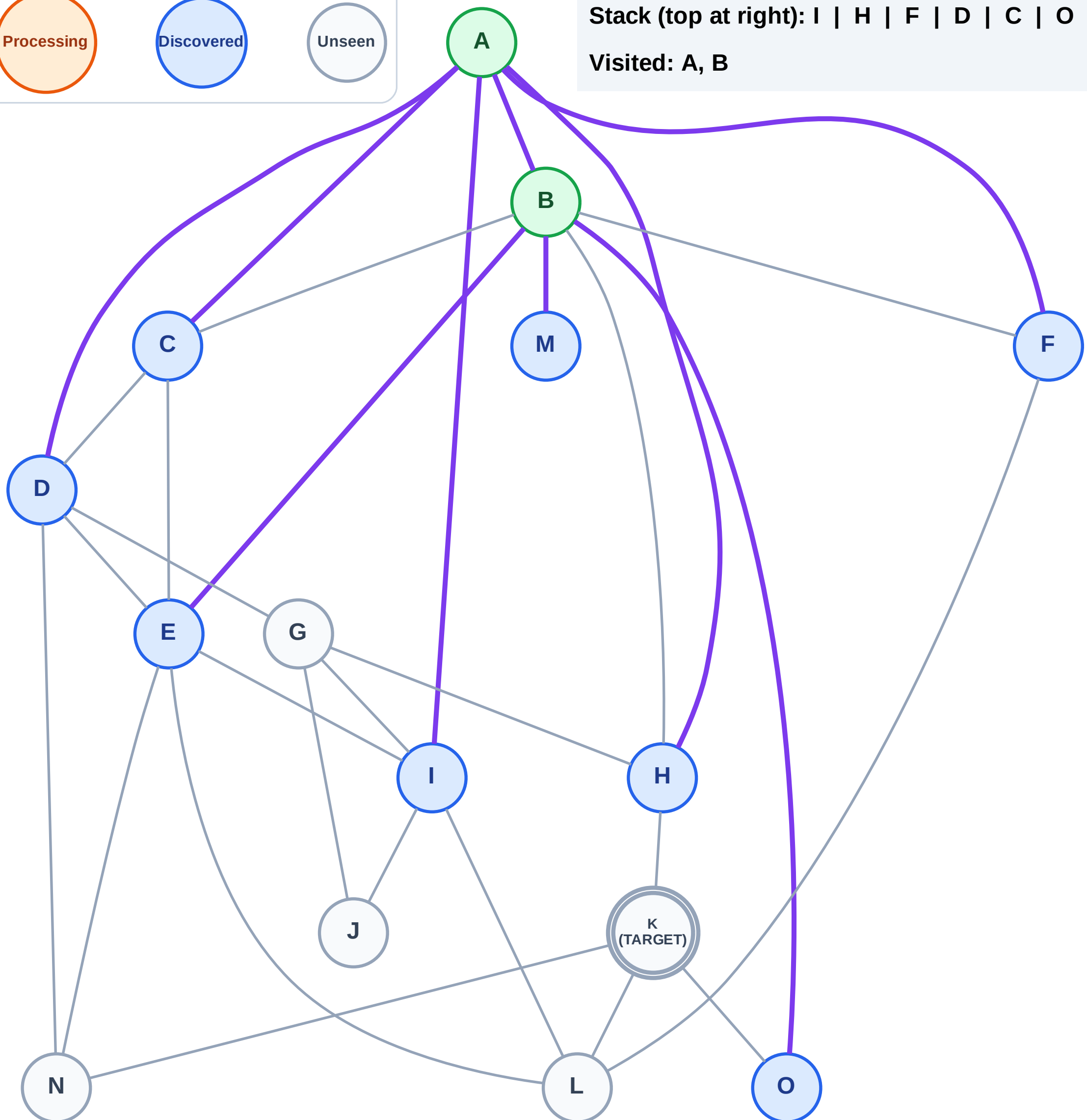
Step 5: Discover O, M, E from B

Legend



Stack (top at right): I | H | F | D | C | O | M | E

Visited: A, B



DFS Traversal

Step 6: Process node E

Legend

Complete

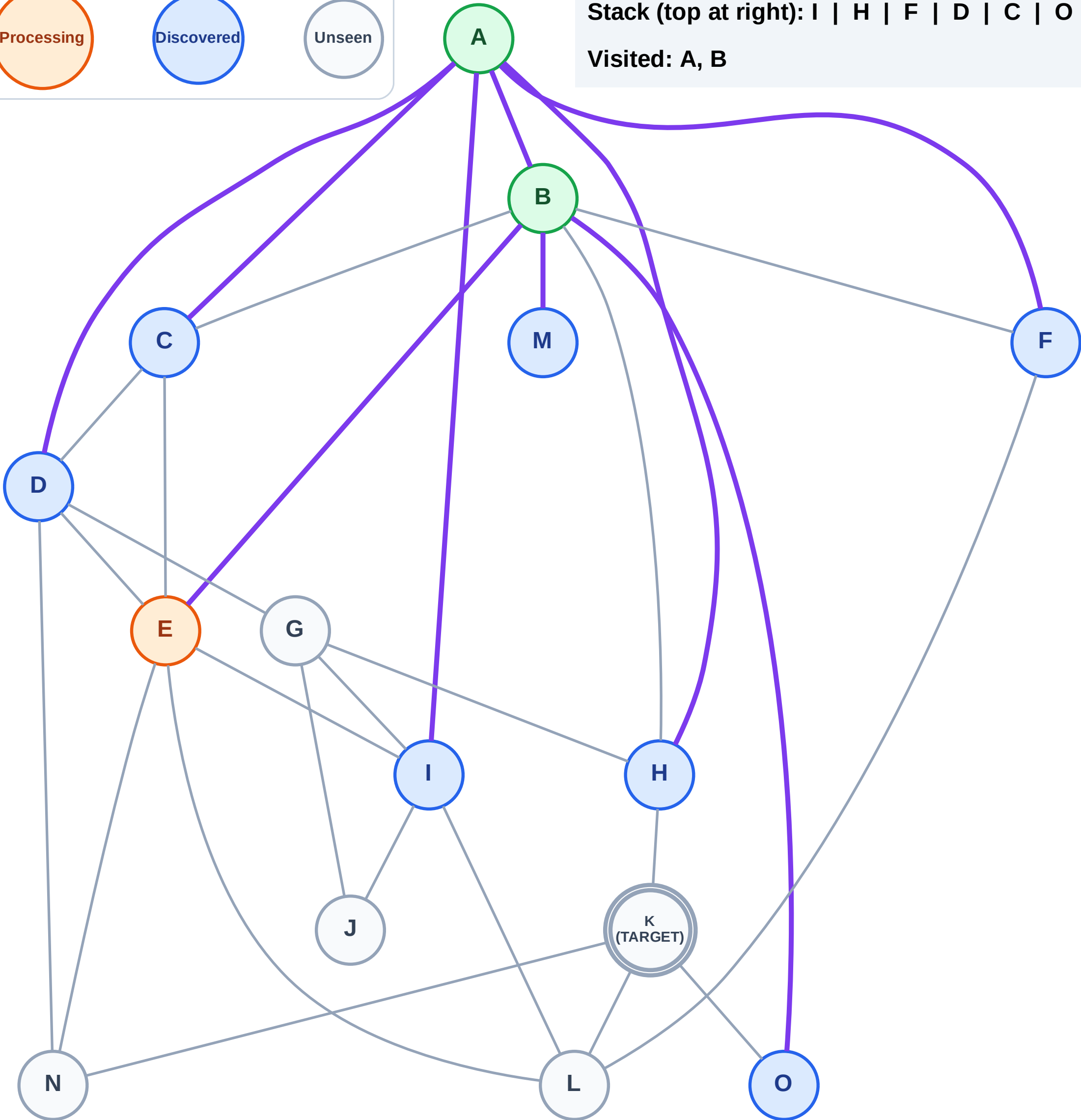
Processing

Discovered

Unseen

Stack (top at right): I | H | F | D | C | O | M

Visited: A, B



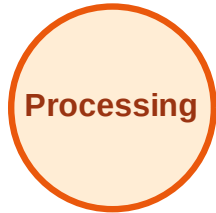
DFS Traversal

Step 7: Discover N, L from E

Legend



Complete



Processing



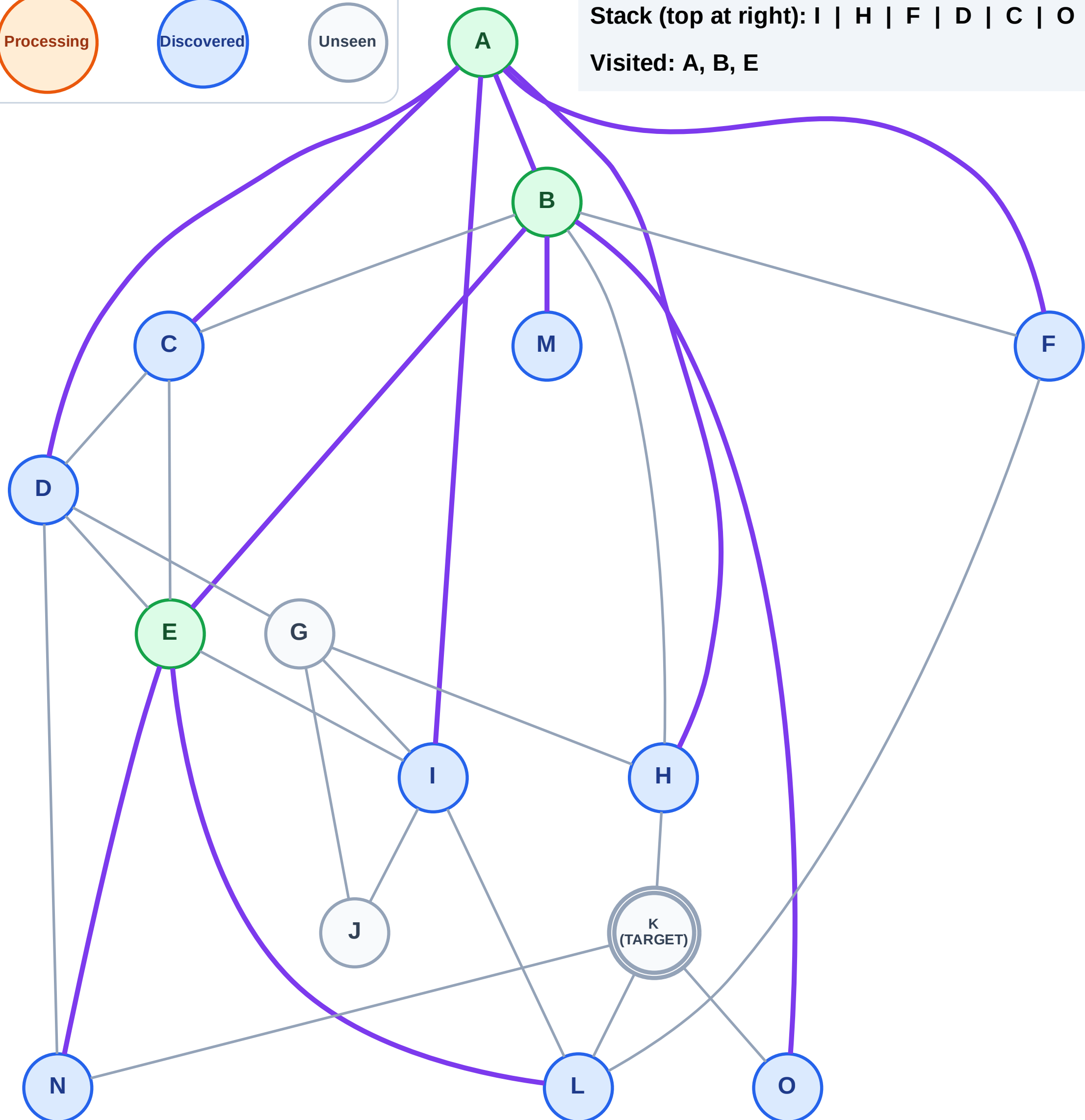
Discovered



Unseen

Stack (top at right): I | H | F | D | C | O | M | N | L

Visited: A, B, E



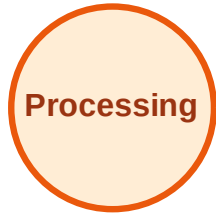
DFS Traversal

Step 8: Process node L

Legend



Complete



Processing



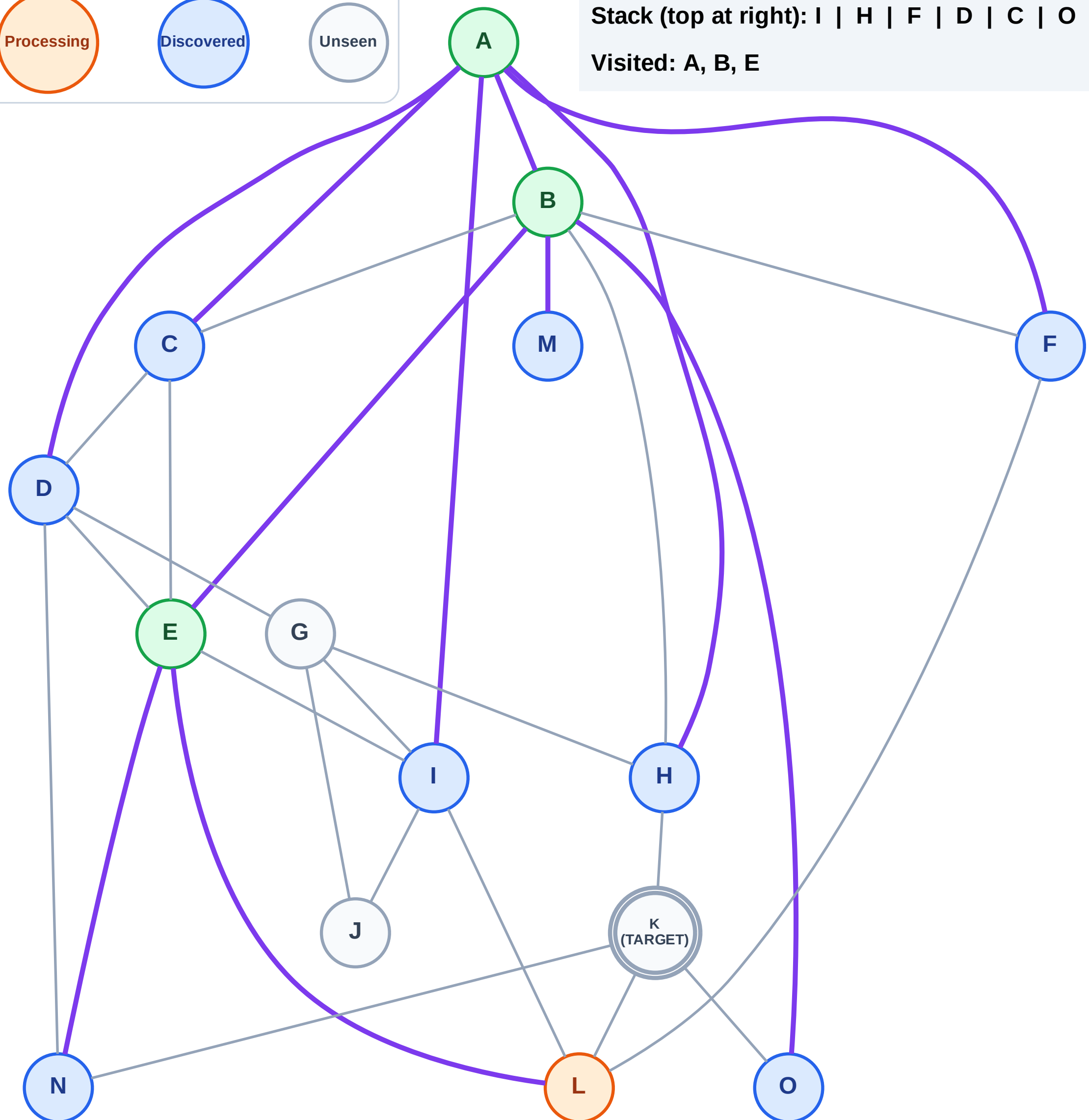
Discovered



Unseen

Stack (top at right): I | H | F | D | C | O | M | N

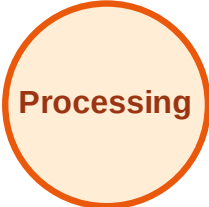
Visited: A, B, E



DFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → E → L → K

Legend



Stack (top at right): I | H | F | D | C | O | M | N

Visited: A, B, E, K, L

